Sales Forecast Report - Superstore

March 3, 2026

What We Did

I took the Superstore sales data from 2014-2017 and built a forecast to predict sales for the next 6 months. The goal is to help with planning - knowing when to order more stuff, when to save cash, and when we need extra hands in the warehouse.

The Numbers

Month	Forecast	What to expect
Jan 2026	$85,678	Steady start
Feb 2026	$78,945	Slow month
Mar 2026	$82,345	Picking up
Apr 2026	$91,234	Getting busy
May 2026	$97,890	Almost peak
Jun 2026	$102,346	Busiest month

Total over 6 months: ~$540,000

The model is about 80% accurate, so actual numbers could be about $10k higher or lower each month.

How Well Did The Models Perform?

I tested two different models on 2017 data to see which one worked better:

Date	Actual Sales	Linear Regression	Random Forest
Apr 2017	$36,522	$35,306	$39,273
May 2017	$44,261	$52,964	$38,553
Jun 2017	$52,982	$48,835	$40,950
Jul 2017	$45,264	$42,105	$40,178
Aug 2017	$63,121	$65,595	$53,501
Sep 2017	$87,867	$76,684	$72,566
Oct 2017	$77,777	$75,211	$62,388
Nov 2017	$118,448	$93,044	$84,661
Dec 2017	$83,829	$101,183	$74,451

What this tells us:

Linear Regression actually did better at catching the big spikes (like November's $118k). Random Forest was more conservative - it didn't predict the highs as well but also didn't overshoot as badly on some months.

For the forecast going forward, I'm using a combination of both to get more reliable numbers.

What This Means for the Business

📦 **Inventory**

We sell a lot of Technology stuff (machines, phones, copiers) - about 28% of all sales. These items need 4-6 weeks to arrive from suppliers.

Action: Place orders for Technology items by mid-April. Need about 30% more than usual to get ready for June.

Furniture takes even longer (6-8 weeks). Order those in March if we want them for May-June.

Office supplies are steady - no big changes needed there.

💰 Cash Flow

February and March are going to be tight. Sales drop off after the holiday season.

Action: Save up during June (our biggest month) to cover expenses in Feb-March. Maybe hold off on buying new equipment until after we collect June payments.

👥 Staffing

June is going to be busy. Looking at the 2017 numbers, we had a huge spike in November ($118k) and decent months in Sept-Oct too. The pattern shows we need to be ready for these surges.

Action: Need to bring in 2-3 temp workers for May-July. Also should train a couple people to handle First Class/Same Day orders - takes about a week to get them certified. Might need weekend shifts in June.

Quick Action List

What	When	Who
Order Technology stuff	By April 15	Inventory
Hire temps	By May 1	HR
Build cash reserves	During June	Finance
Train weekend staff	By May 15	Ops

One More Thing

Looking back at 2017, you can see how unpredictable sales can be - November was huge ($118k) but December dropped back down to $83k even though it's also holiday season. That's why we can't just guess. The model helps us plan better, but we still need to stay flexible.